Training datasets and why each one needs a licence

DFU Thermography Screening Station (DFUSCAN-01) · eMathrix Technologies · Revised November 12, 2026

This document explains which public datasets can be used to train the station's image classifier, and why the licence of every dataset has to be checked before it is used. It is written for the research team and the project manager. It is an engineering checklist, not legal advice: for the final decision, ask the university's research office or a legal adviser.

> **The model.** The classifier is MobileNetV2 (objective 2). It looks at the **colour** photograph of the soles and outputs a predicted class. It is trained on a separate computer, exported to ONNX and compiled to a `.hef` file that runs on the Hailo-8L of the AI HAT+. Wagner grades used as training labels come only from a clinician; the station never assigns a grade. The 2.2 °C left–right temperature check is not a model: it is fixed arithmetic and needs no training.

1. The datasets supplied so far

Dataset	What it contains	Fit for the colour classifier
ThermoDataBase (IEEE DataPort)	Thermal images and temperature tables of both feet, 45 control and 122 diabetic subjects. Labels: diabetic or control only.	**No: thermal, not colour; no Wagner grades. Used instead to test the 2.2 °C rule (section 2).**
Thermography images of diabetic foot (Kaggle)	A copy of ThermoDataBase pictures, rotated and sharpened, split into train and val.	**No: same images; 2 patients appear in both train and val, which inflates accuracy.**
DFU Wagner's classification (Kaggle, purushomohan)	Colour photos in folders by grade.	**Yes for grades 0–2, after the licence and folder names are checked.**
DFU dataset annotated into 4 classes (Kaggle, khalidsiddiqui2003)	Colour images in 4 classes.	**Only if the classes are Wagner grades. If they are infection / ischaemia / both / none (the DFUC2021 scheme), no.**
Diabetic foot ulcer (Kaggle, laithjj)	Small patches of healthy skin or ulcer.	**Not as grade 0 unless the study clinician confirms it in writing: healthy skin is not the same as a Wagner 0 at-risk foot.**
Multimodal DFU Classification (Mermer, Aksu et al.; authors' GitHub, re-uploaded to Kaggle by prantadhar)	RGB (Pi Camera V2) and thermal (FLIR Lepton 3.5) pictures from a Raspberry Pi device, Wagner 0–5, graded by experts.	**Yes for grades 0–2, as a stand-in for the prototype only. No licence: see section 3.**

One more risk: when each grade comes from a different dataset, a model can learn to recognise the dataset (camera, light, crop) instead of the grade, and still score well. Published checks of Kaggle diabetic-foot datasets also report mislabelled and duplicated images. The training pipeline removes exact duplicates and keeps near-duplicates of one foot together, but this limitation must be stated with any result.

2. What the thermal database showed

The station's own asymmetry code (foot measurement, 8 regions, inclusive 2.2 °C rule) was run on all 167 subjects of ThermoDataBase. Nothing was trained.

	Control	Diabetic
Subjects analysed	45	117 (5 not measurable)
At least one region flagged (≥ 2.2 °C)	4 (9 %)	28 (24 %)
Largest regional difference, median	0.88 °C	1.37 °C

Whole-foot means matched the database's own values exactly, and the left/right proposal from foot shape agreed with the database's labels in 160 of 160 clear cases. A flag is a temperature difference, not a diagnosis; the database has no lesion or grade labels, and it was taken with another camera and protocol.

3. The multimodal Raspberry Pi dataset

Source: Mermer G., Aksu M. F., Ozsezer G., Cetinkalp S., Er O., Gullu M. K., *Multimodal Deep Learning for Diabetic Foot Ulcer Staging Using Integrated RGB and Thermal Imaging*, arXiv 2603.26952 (2026). Images: github.com/faksu267/Multimodal-Diabetic-Foot-Ulcer-Classification. This is the closest public match to the station found so far: a Raspberry Pi device with a Pi camera and a FLIR Lepton 3.5.

Wagner grade	RGB images	Separate capture sessions (approx.)
0	150	6
1	106	15
2	496	30
3, 4, 5 (not used)	226, 184, 43	8, 7, 5

Sessions are counted from the file timestamps: images less than 2 minutes apart are counted as one session.

- **Burst shots.** The pictures were taken in bursts seconds apart, on 4 days. The real number of patients is much smaller than the number of images. The paper's 93 % accuracy used a split that is not grouped by patient, so frames of the same foot can sit on both sides. The station's pipeline groups near-identical frames before splitting, so its accuracy will be lower. That lower number is the honest one.
- **No licence.** The authors' GitHub repository has no licence file and asks only that the paper be cited. The Kaggle copy was uploaded by someone who is not an author and is marked "Unknown". Without a licence, all rights stay with the authors.
- **Ethics.** The patient consent and ethics approval are not stated in the repository.

> **Conditions for using it in the prototype (November 18 demonstration):** (1) email the authors for permission for academic use and keep the email; (2) download from the authors' GitHub, not the Kaggle copy; (3) cite the paper; (4) tell the panel it is a stand-in dataset, permission is requested, and the model will be retrained on the team's own data after ethics approval; (5) it is not the final model of the delivered station.

4. How many images per grade

What counts is the number of **different patients or feet**, not the number of pictures. Ten shots of one foot taken seconds apart count as one foot.

	Images per grade	Different patients per grade
Minimum	about 150	20–30

Better	300 or more	50 or more

Below about 100 images or a handful of patients per grade, training still runs but the result is unstable and must be stated as a limitation. If the minimum is not reached:

1. Combine sources: clinician-graded Mendeley images plus the multimodal dataset (grades 0–2).
2. Reduce the classes to two: grade 0 against grades 1–2. This is a valid scope and needs less data.
3. Augmentation (flip, brightness, contrast) is already part of training. It does not count as new data.
4. Never copy files or relabel images to raise the count. Report the real numbers.

Selecting Mendeley images

- Plantar (sole) views only, with most of the sole visible, as the station photographs the soles from below.
- Leave out top-of-foot, side and leg views, close-ups of the wound alone, blurred or dark pictures, and feet covered by paper, fingers, hands or bandages.
- Only the study clinician assigns the grade. If the clinician is unsure, leave the image out.
- Keep the original files unedited and send a CSV with `filename, wagner_grade, clinician, date_graded, notes`. Note in `notes` when several images show the same patient.

5. Other sources searched

Source	Labels	Licence	Use
Lower Limb and Feet Wound Image Dataset (Mendeley Data, DOI 10.17632/hsj38fwnvr.2)	Normal / wound, 8,129 images	CC BY 4.0	**Only if the study clinician assigns the Wagner grades**
DFUC2021 (Manchester Metropolitan University)	Control, infection, ischaemia, both	On request	**No: not Wagner**
Published studies with Wagner-graded photos	Wagner grades	Not published	**Only by asking the authors**

Apart from the multimodal dataset in section 3, which has no licence, no openly licensed dataset was found with colour photographs labelled Wagner 0, 1 and 2. The two cleanest routes are: (a) an openly licensed image set graded by the study clinician, and (b) scans from the station itself, graded by the clinician. Route (b) is the only one that matches what the station sees.

6. Why every dataset needs a licence

A dataset is a component

Every part in the station's bill of materials has a datasheet and a supplier. A dataset is a component of the model in the same way, and needs the same checks.

Hardware part	Dataset
Datasheet: ratings, pinout	Description: what the images show, the labels, who labelled them
Supplier and purchase terms	**Licence**: what may be done with the images
Counterfeit part from an unknown seller	Images re-uploaded by someone who does not own them
Part number and lot traceability	Address, licence and download date kept with the data

Every step of the data flow is a use the licence must allow

```
Images ──copy──► Training PC ──train──► Weights ──compile──► .hef ──deliver──► Station at client
         (1)                    (2)               (3)               (4)                (5)
```

Step	Question for the licence
1. Copy	May the images be downloaded and stored?
2. Modify	May they be resized, cropped and augmented?
3. Train	May a model be derived from them? The weights are learned from the images.
4. Distribute	May the trained model be given to someone else?
5. Commercial use	The station is delivered under a paid contract (EMX-2026-DFUSCAN-01). Is that allowed?

If any step is not allowed, the whole chain fails. A non-commercial licence can be acceptable for thesis experiments but not for the model inside a station delivered under a contract.

Common licences

Licence	Train a model	In the delivered station	Obligation
CC0 / public domain	Yes	Yes	None (citing is still good practice)
CC BY	Yes	Yes	Credit the authors in the thesis and documentation
CC BY-SA	Yes	Unclear	Derived material may have to be shared under the same licence
CC BY-NC	Research only	No	Non-commercial use only
CC BY-ND	Unclear	Unclear	No modified versions; augmentation is a modification
Agreement on request (e.g. DFUC)	As agreed	Usually no	Sign and follow the agreement
No licence stated	No	No	All rights stay with the owner by default

Provenance: who really owns the images

On Kaggle the uploader chooses the licence label, even for images that are not theirs. If a group's images are re-uploaded, the original owner's terms still apply, whatever the Kaggle page says. A licence passes down the supply chain the way a counterfeit part stays counterfeit after it is resold.

Medical images carry patient rights

- **Consent and ethics.** Foot photographs come from patients who agreed to a specific use, usually under an ethics approval for a stated purpose.
- **Data Privacy Act (RA 10173).** Health information is sensitive personal information in the Philippines. This is also why the station stays offline and stores only reference codes. Images of unknown origin bring that risk back into the project.

Traceability for the thesis and the client

The evaluation in objectives 4 and 5 is only defensible if every training image can be traced to its source and its permission. The training configuration already enforces this: every dataset entry must state its address, licence, download date and who confirmed the label mapping, and the intake report prints them. A dataset folder with no entry is refused.

What can go wrong without a clear licence

Event	Effect on the project
The owner asks for the images to be removed	The model must be retrained without them; every reported result changes
The licence turns out to be non-commercial	The delivered station's model is in breach
The images turn out to be re-uploads	Thesis results can be challenged
Patient consent did not cover this use	Ethics and privacy exposure for the school and the client

Decision for each dataset

```
Licence stated? ── no ──► do not use
      │ yes
Uploader owns the images, or they trace to the original? ── no ──► do not use
      │ yes
Non-commercial (NC)? ── yes ──► thesis experiments only, not in the delivered station
      │ no
Credit required (BY)? ── yes ──► use, and cite in the thesis and documentation
      │
Use, and record the address, licence and date in training/configs/public_v1.yaml
```

7. Next steps

1. Research team: email the authors of the multimodal dataset for permission, today.
2. Research team: select plantar Mendeley images and have the clinician grade them; send the images and the CSV.
3. eMathrix: download grades 0–2 of the multimodal dataset from the authors' GitHub, run the intake and the near-duplicate grouping, and train the prototype model for November 18.
4. Add the graded Mendeley images when they arrive and retrain.
5. December–March, after ethics approval: collect graded scans with the station and retrain for the final defense.
